

Four-Level SapBERT: Model Comparison

Revised comparison | Levels 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent

This improvement study uses the same 10,000 supplied records, frozen SapBERT embeddings and original grouped partitions. It expands the search to PCA-64, PCA-128 and PCA-256, stronger/weaker regularization, class balancing, PCA whitening and structured-only controls. No labels are changed.

Selected: SapBERT + PCA-128 + Logistic Regression

The new CatBoost, XGBoost and SVM trials did not beat this configuration. The existing model is retained; the 90% target across all four aggregate metrics remains unmet.

Training settings: C=10; balanced class weights.

Patient preprocessing includes quadratic numeric terms. SapBERT remains frozen.

An additional 19 explicit detail features preserve severity, onset, duration and other mentions from the original complaint alongside the SapBERT concept embedding.

Selected total input size: 197 features. PCA retained variance: 95.51%.

Retrospective metric (%)	Initial	Prior round	Selected
Accuracy	85.19	89.29	89.29
Macro precision	85.71	89.67	89.67
Macro recall	85.90	89.81	89.81
Macro F1	85.80	89.73	89.73
Emergency recall	92.16	95.10	95.10
Under-triage	7.60	4.65	4.65

Selected quadratic weighted kappa: 0.9504; mean absolute level error: 0.1081; over-triage: 6.05% (initial 7.20%).

Selected retrospective MCC: 0.8563; log loss: 0.2419; multiclass Brier score: 0.1497.

Mean development CV macro F1 change versus initial: +3.67 percentage points. Selection uses all five development folds and an emergency-recall constraint; the test results above do not select the winner.

Paired development bootstrap versus initial: F1 gain 95% interval +2.97 to +4.33 percentage points. This conditional interval excludes zero. This diagnostic excludes model-selection uncertainty.

Evaluation scope: These 1,999 test rows were already examined in the previous study. The new scores are retrospective comparisons, not a fresh independent estimate. New labelled records are needed to confirm generalization. Saved concepts are evaluated here; live translation accuracy is not measured.

Trade-offs remain visible: compare emergency recall and under-triage as well as aggregate scores. SapBERT is not fine-tuned; these gains do not establish clinical validity.

Complete Development Cross-Validation

All 79 configurations; 395 fits across five grouped folds. F1, accuracy and emergency recall are percentages. SD is the fold-to-fold F1 standard deviation in percentage points. The original reference is lr_pca64_c10_balanced1.

Configuration	F1	SD	Accuracy	Emergency recall
lr_detail128_c10	89.65	0.26	89.28	94.06
lr_detail64_c100	89.56	0.82	89.19	93.93
logreg_complaint_details	89.38	0.50	88.99	93.63
lr_detail128_c30	89.31	0.38	88.91	93.93
lr_detail128_unbalanced	89.30	0.33	88.89	93.01
lr_detail128_c1000	89.08	0.52	88.66	93.26
lr_detail128_c300	89.00	0.57	88.58	93.44
lr_detail128_c1	88.68	0.63	88.29	94.42
lr_detail256_c100	88.65	0.36	88.21	93.01
hgb_complaint_details	88.07	0.95	87.65	92.89
logreg_concept_details	87.83	1.06	87.40	93.38
hgb_detail64_leaf7_l220	87.68	0.73	87.25	92.77
cat_detail64_depth6	87.23	0.70	86.88	93.26
cat_detail64_depth4	86.86	0.69	86.51	94.24
hgb_detail64_leaf15_l210	86.86	0.83	86.40	91.73
lr_quad128_c100	86.57	0.70	86.09	92.65
lr_pca128_c100_balanced0	86.57	0.59	86.05	91.60
lr_refine128_c300	86.50	0.75	86.00	92.52
lr_pca128_c1000_balanced1	86.48	0.69	85.98	92.28
lr_quad128_c300	86.48	0.68	85.99	92.34
lr_pca128_c100_balanced1	86.46	0.58	85.98	92.65
lr_quadratic64_c100_balanced1	86.41	0.80	85.93	92.83
xgb_detail64_depth4	86.39	0.83	85.95	91.85
lr_pca128_c10_balanced0	86.39	0.57	85.84	91.73
lr_refine128_c30	86.32	0.51	85.81	92.89
lr_quadratic64_c100_balanced0	86.31	0.51	85.81	91.05
cat_detail128_depth6	86.30	0.65	85.88	93.81
lr_refine64_c30	86.26	0.61	85.75	93.50
lr_pca128_c1000_balanced0	86.24	0.47	85.71	91.05
lr_pca256_c10_balanced0	86.24	0.44	85.68	91.67
lr_quadratic64_c10_balanced0	86.23	0.66	85.71	91.24
lr_quadratic64_c10_balanced1	86.21	0.52	85.71	93.08
lr_pca64_c1000_balanced0	86.19	0.54	85.68	90.93
lr_pca64_c100_balanced1	86.19	0.46	85.69	93.01
lr_quad256_c100	86.18	0.92	85.66	91.54

Development Cross-Validation: Continued

Configuration	F1	SD	Accuracy	Emergency recall
lr_pca64_c100_balanced0	86.17	0.78	85.65	91.05
lr_pca128_c10_balanced1	86.16	0.43	85.63	93.07
lr_refine64_c300	86.06	0.52	85.55	92.83
lr_pca256_c100_balanced0	86.06	0.57	85.50	90.75
lr_pca256_c10_balanced1	86.05	0.46	85.51	92.95
lr_pca64_c1000_balanced1	86.04	0.55	85.53	92.77
hgb_concept_details	86.02	1.03	85.54	91.48
lr_pca64_c10_balanced1	85.98	0.53	85.46	93.38
cat_detail128_depth4	85.94	0.78	85.56	94.24
xgb_detail64_depth6	85.91	0.82	85.50	91.18
lr_pca256_c100_balanced1	85.91	0.70	85.36	91.97
lr_quad256_c300	85.90	1.07	85.35	91.24
lr_pca64_c10_balanced0	85.86	0.61	85.31	91.48
hgb_detail64_leaf31_l220	85.85	0.45	85.40	91.12
xgb_detail128_depth4	85.85	0.92	85.36	92.89
svc_detail128_c10	85.49	0.34	85.04	89.22
xgb_detail128_depth6	85.42	0.85	84.98	92.16
svc_detail128_c1	85.20	0.29	84.70	88.66
lr_pca256_c1000_balanced1	85.14	0.76	84.54	90.99
lr_pca256_c1000_balanced0	85.12	0.67	84.50	90.32
lr_pca128_whiten	85.06	0.46	84.51	92.34
lr_pca64_whiten	85.04	0.57	84.54	92.09
hgb_refine64_leaf7	84.76	0.80	84.26	90.38
hgb_fused_64_leaf7	84.34	0.65	83.81	91.36
hgb_fused_64_leaf15	84.24	0.76	83.74	90.81
svc_detail128_c100	84.24	0.48	83.74	87.07
hgb_fused_128_leaf15	83.75	0.59	83.23	90.44
lr_pca256_whiten	83.49	0.82	82.89	90.56
hgb_refine64_leaf31	83.45	0.63	82.94	90.14
hgb_fused_128_leaf7	83.41	0.98	82.83	91.61
lr_quadratic64_c1_balanced1	83.21	0.55	82.64	93.02
lr_quadratic64_c1_balanced0	83.20	0.73	82.56	90.26
rf_detail64_fraction0.3	81.96	0.63	81.58	91.79
rf_detail64_fraction0.7	81.55	0.88	81.10	88.73
rf_complaint_details	80.32	0.79	80.24	95.89
rf_concept_details	80.00	0.57	79.93	96.32
rf_refine64_leaf1	79.21	0.34	79.13	95.96
rf_refine64_leaf3	78.31	0.42	78.33	96.63
rf_fused_64	78.31	1.21	77.82	87.68
rf_refine128_leaf3	76.48	0.76	76.72	96.63
rf_refine128_leaf1	76.48	0.85	76.64	95.77
hgb_structured_0_leaf7	76.23	1.18	75.55	87.38
hgb_structured_0_leaf15	75.87	1.00	75.15	86.40
rf_structured_0	75.50	0.73	74.85	86.03

Selection: highest mean macro F1 among candidates whose mean emergency recall is within one percentage point of the original reference; accuracy resolves ties. A higher F1 does not qualify a candidate whose emergency recall falls below the threshold. This is not a clinical safety guarantee.

Original 768-D versus PCA-64 Baselines

All original fixed comparisons are retained below. They use the same 1,999 previously examined test rows. Values are percentages; precision, recall and F1 are macro-averaged. These baseline runs are from the first round and are not new experiments.

Text only

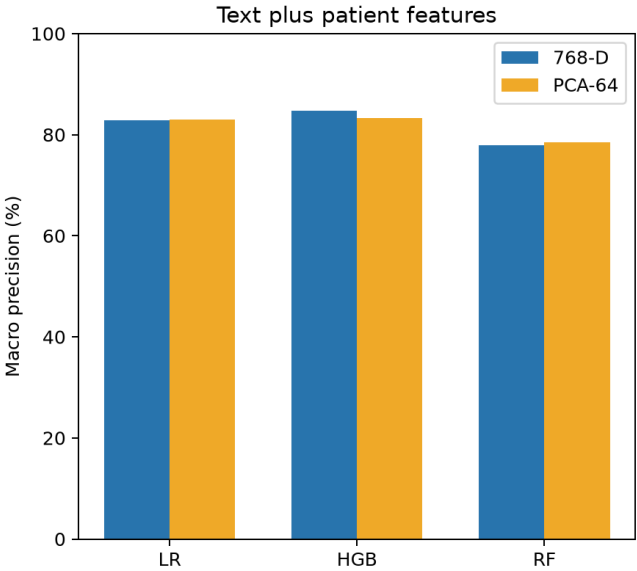
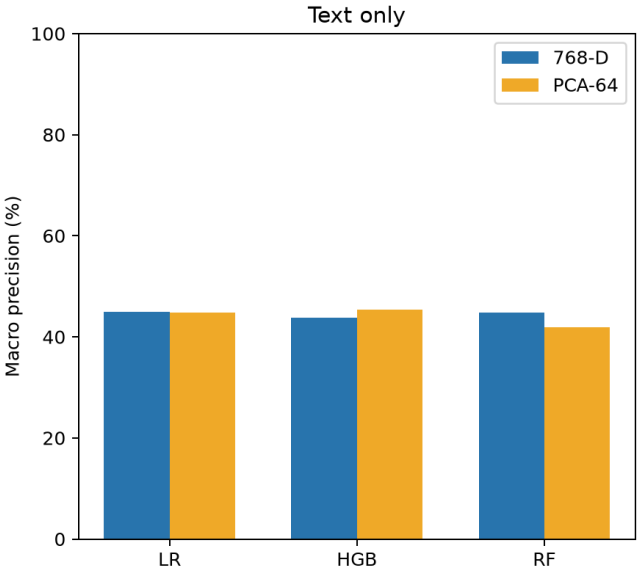
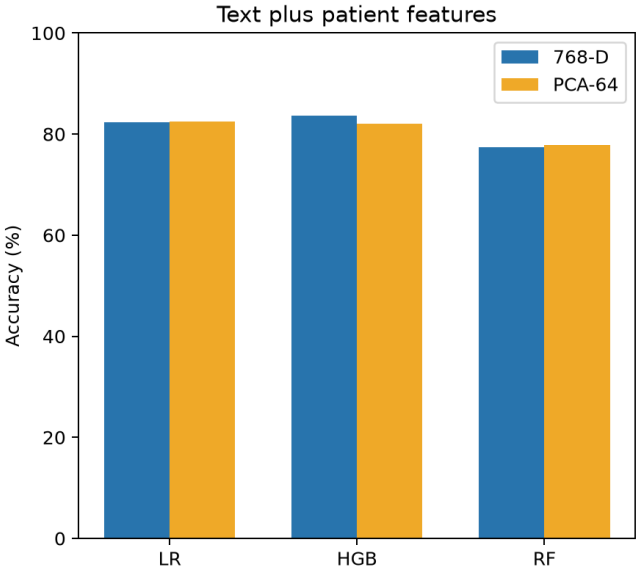
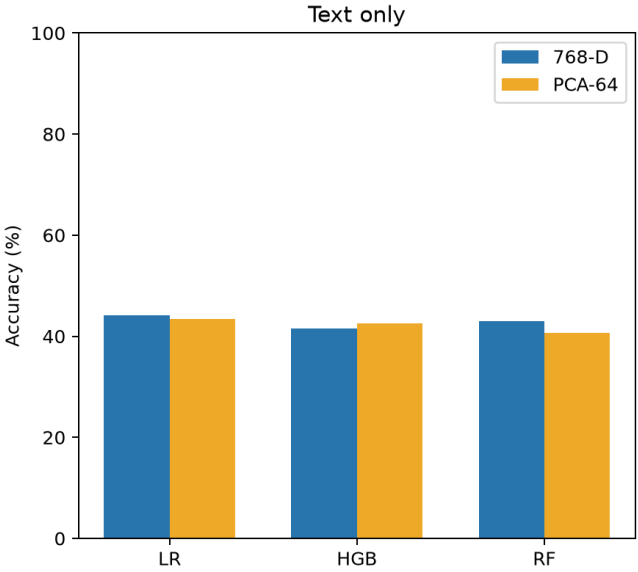
Text size	Classifier	Accuracy	Precision	Recall	F1
768	Logistic Regression	44.17	44.89	44.87	43.95
768	HistGradientBoosting	41.52	43.72	40.95	41.70
768	Random Forest	42.92	44.82	42.93	43.49
64	Logistic Regression	43.42	44.74	44.20	43.26
64	HistGradientBoosting	42.57	45.34	41.85	42.80
64	Random Forest	40.62	41.88	40.93	41.08

Text plus patient features

Text size	Classifier	Accuracy	Precision	Recall	F1
768	Logistic Regression	82.29	82.89	83.10	82.92
768	HistGradientBoosting	83.64	84.72	84.03	84.21
768	Random Forest	77.44	77.87	78.24	77.93
64	Logistic Regression	82.44	83.01	83.22	83.03
64	HistGradientBoosting	82.09	83.28	82.54	82.76
64	Random Forest	77.89	78.47	78.65	78.35

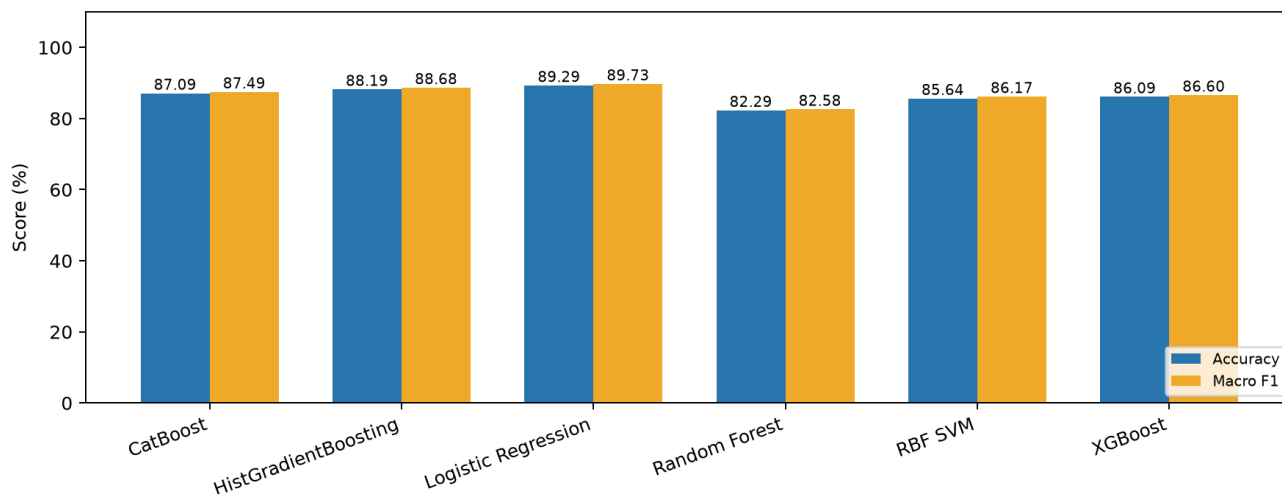
The original fused PCA-64 input has 86 features: 64 text components and 22 encoded patient features. The full fused input has 790 features. All training transformations are fitted without validation/test rows.

Performance Graphs



Expanded Classifier Comparison

Tuned classifiers: retrospective comparison

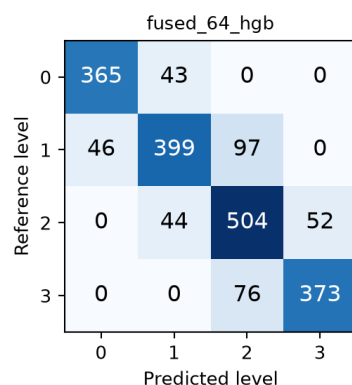
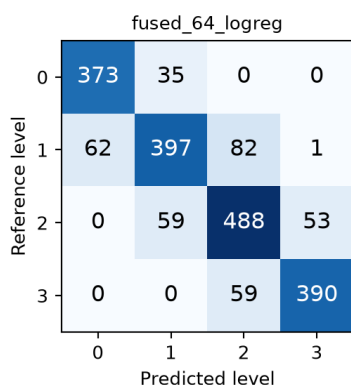
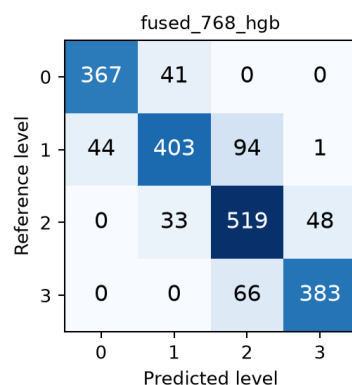
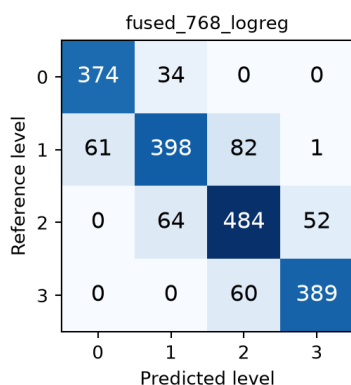
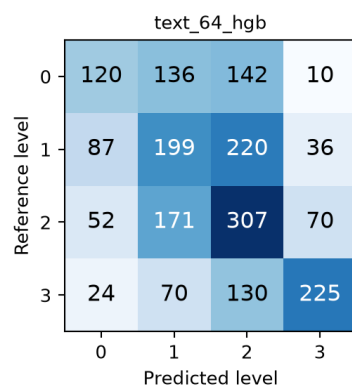
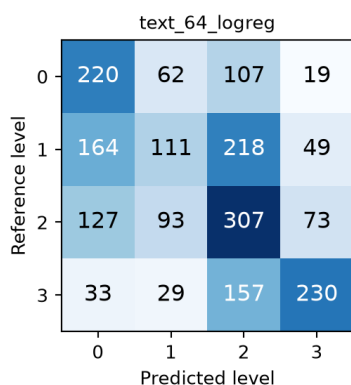
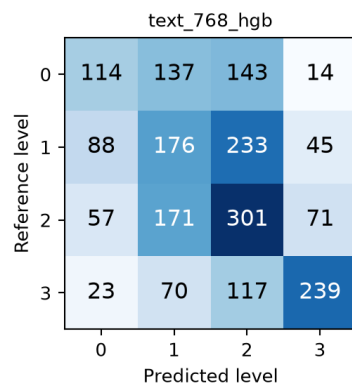
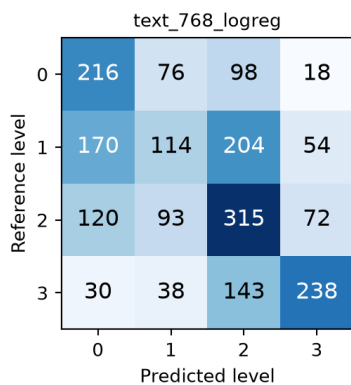


Classifier	Accuracy	Precision	Recall	F1
CatBoost	87.09	87.53	87.50	87.49
HistGradientBoosting	88.19	88.86	88.58	88.68
Logistic Regression	89.29	89.67	89.81	89.73
Random Forest	82.29	83.00	82.76	82.58
RBF SVM	85.64	86.38	85.99	86.17
XGBoost	86.09	86.71	86.54	86.60

Each family uses its highest fused development CV F1 setting. These descriptive choices do not override the emergency-recall constraint for deployment. Scores are percentages; precision, recall and F1 are macro averages.

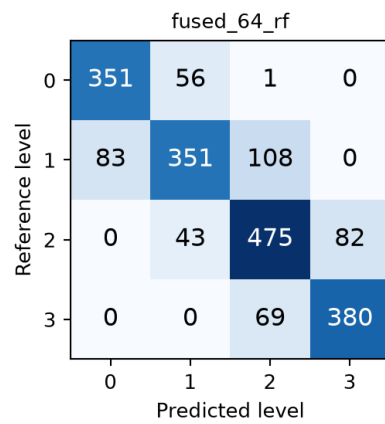
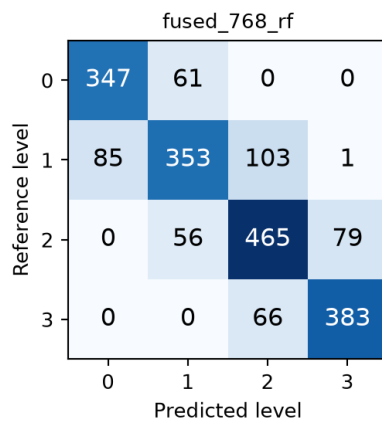
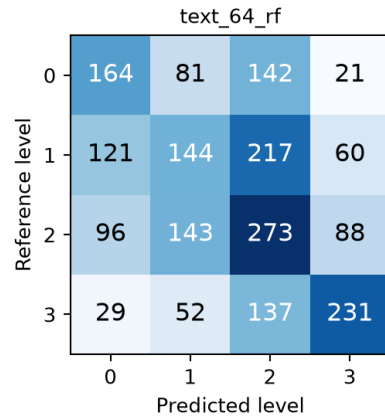
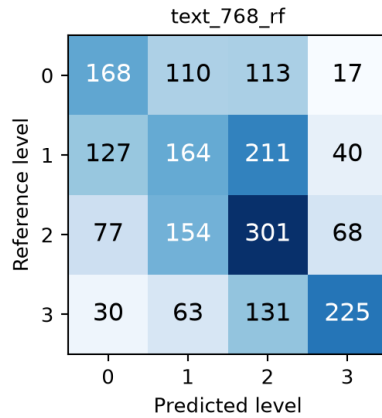
Blue and yellow bars preserve the original comparison style. The selected improved result is shown separately on page 1 because it follows a larger development search.

Baseline Confusion Matrices: LR and HGB



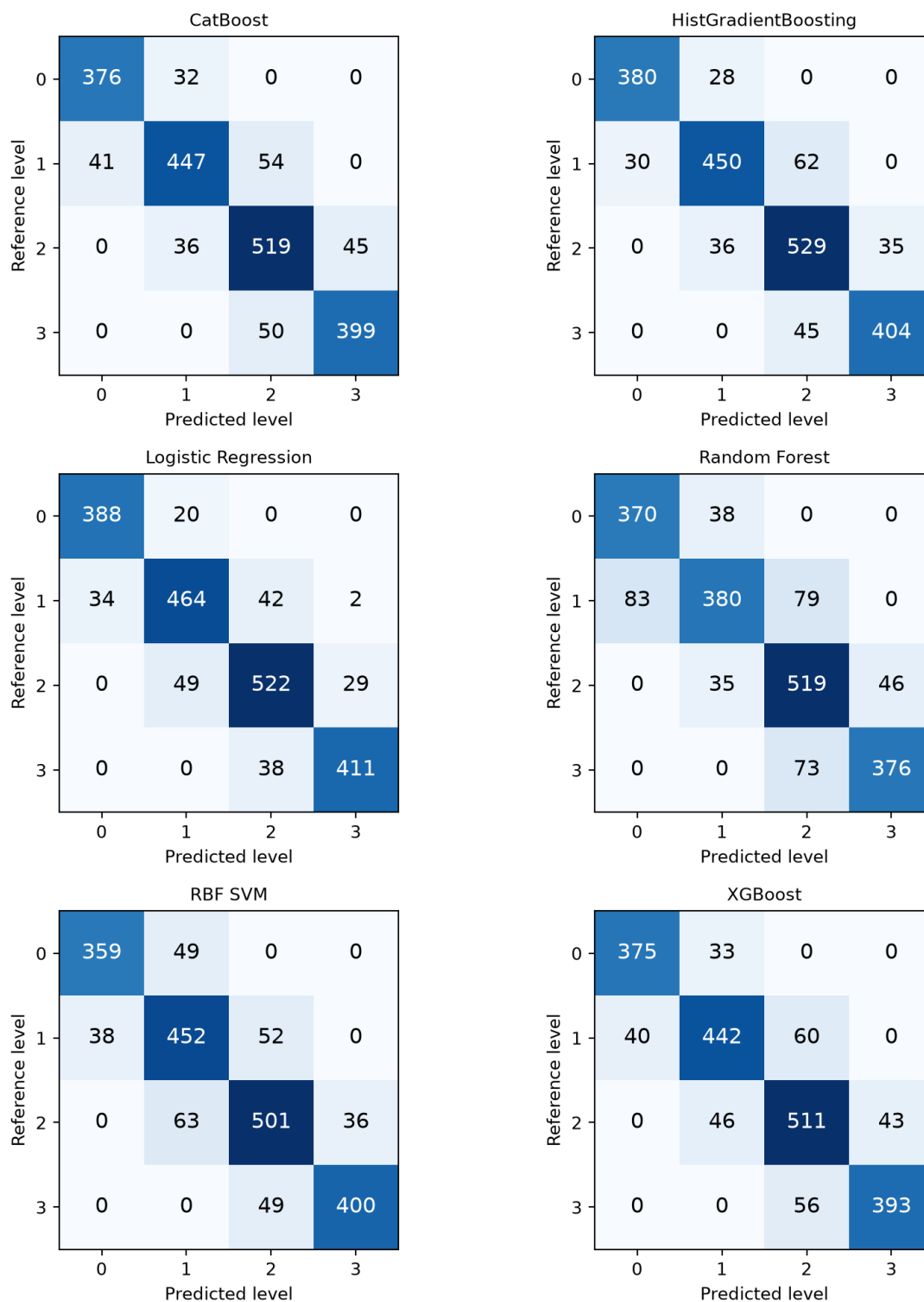
Rows are reference labels; columns are predictions. Level order: 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent. Baseline matrices are unchanged from round one.

Baseline and Selected Confusion Matrices



Rows are reference labels; columns are predictions. Level order: 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent. Baseline matrices are unchanged from round one.

Tuned Classifier Confusion Matrices



All matrices contain the same 1,999 records. Rows are reference labels and columns are predictions, in level order 0, 1, 2, 3. Family representatives are selected by development scores.

Class Results and Error Review

Level	Precision (%)	Recall (%)	F1 (%)	Support
0 Emergency	91.94	95.10	93.49	408
1 Urgent	87.05	85.61	86.33	542
2 Standard	86.71	87.00	86.86	600
3 Non-urgent	92.99	91.54	92.26	449

Development out-of-fold confusion matrix

Reference / predicted	0	1	2	3
0	1535	97	0	0
1	141	1851	177	3
2	0	162	2090	148
3	0	0	130	1667

An input audit found no identical development input combinations with conflicting targets. This does not establish that labels are clinically correct. A local review CSV lists development errors and model confidence; it is not committed because it contains individual records.

Use the review list to investigate unclear label boundaries, missing symptom severity/duration/history and data entry issues. Do not replace reference labels with model predictions merely to improve agreement.

Literature context (different task)

Model	Accuracy (%)	Recall (%)	F1 (%)
TF-IDF + LR	75.00	98.80	54.40
TF-IDF + RF	75.10	96.40	58.20
BiLSTM	74.60	84.60	67.00
CNN	72.30	78.70	66.70
Current four-level SapBERT	89.29	89.81	89.73

Seo et al., Scientific Reports (2025), Table 2. DOI: 10.1038/s41598-025-99874-0. Different binary KTAS task; publication metric definitions differ. Context only, not evidence of superiority over these models.

Methods, Reproducibility and Limits

Data and preprocessing

The original 8,001 development / 1,999 test partition and five grouped folds are reused. Groups do not overlap across training and validation/test boundaries. Input and embedding hashes are checked before and after the run. Median imputation, categorical encoding, scaling and PCA are learned inside each development fold. No label-derived columns are model inputs.

Encoder and search

SapBERT-from-PubMedBERT-fulltext uses frozen 768-D CLS embeddings, L2 normalization, a 64-token limit and the original pinned revision. Logistic Regression varies $C=10$, 100 and 1000 with and without balancing, plus whitened PCA with $C=0.1$. Six additional LR configurations use quadratic numeric terms, with $C=1/10/100$ and optional balancing. HGB tests 7/15 leaves with regularization; Random Forest tests structured and fused inputs. Every candidate is evaluated on all five folds.

The refinement adds 14 configurations: balanced LR $C=30/300$ at PCA-64/128; quadratic LR $C=100/300$ at PCA-128/256; HGB with 7/31 leaves and 800/400 iterations; and RF with 500 trees, square-root feature sampling and leaf sizes 1/3. These historical refinement settings retained the original three classifier families.

Six additional settings compare 19 explicit details extracted from English concepts or original complaints, using each family's prior best setting. The separate learning-curve audit uses nested whole-group subsets; labels remain unchanged.

Selection and deployment

The expanded comparison adds eleven settings: CatBoost depth 4/6 with 800 iterations and XGBoost depth 4/6 with 600 trees, each at PCA-64/128; RBF SVM $C=1/10/100$ at PCA-128. All retain original-complaint detail features. New dependencies and parameters are recorded with the experiment.

The incumbent is included in the same search. Selection is recorded before the retrospective test evaluation. The exported model must reproduce saved predictions through the shared GUI/CLI preprocessing. Model artefacts and the incumbent are retained separately. Quadratic terms, if selected, are applied by the fitted numeric preprocessing pipeline. No label correction or encoder fine-tuning is claimed.

Research interpretation

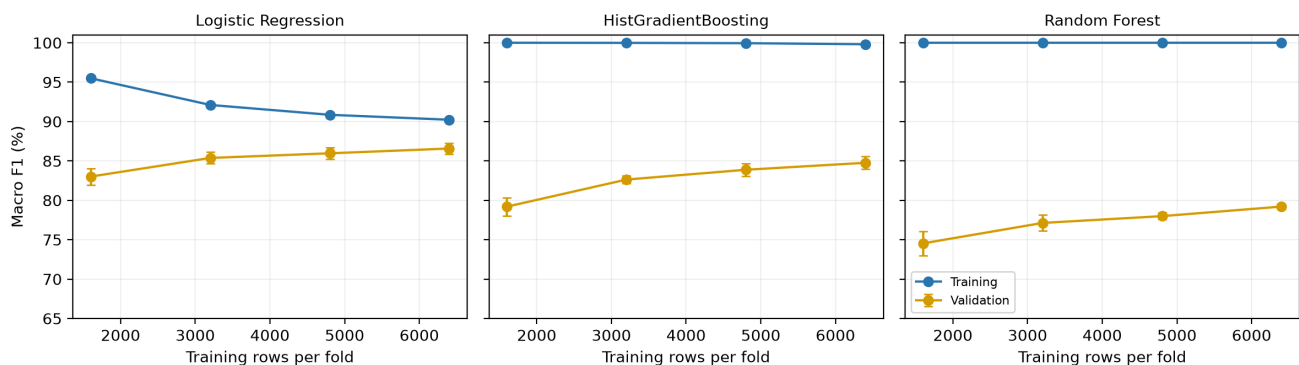
The larger search can overfit development cross-validation. The old test set is already exposed, so its scores must not be presented as untouched confirmation. A new independently labelled dataset is needed for confirmation. Supplied/recovered concepts bypass live translation during this evaluation. The provider could not supply label-assignment rules; independent per-record review is not documented. These are research comparisons, not clinical validation.

Files

protocol.json freezes candidates and source identities; cross_validation.csv contains every fit; cv_summary.csv reports every candidate; selection.json records the decision; retrospective_results.json contains class scores and matrices. Source data, embeddings and per-record error files remain local.

Complaint Audit, Learning Curves and Next Steps

The development audit flags 567 explicit duration disagreements between original complaints and supplied concepts, 142 durations not recovered from concepts and 48 family-word omissions. These lexical flags require review; they do not establish incorrect clinical labels. Labels remain unchanged.



Nested whole-group subsets use 25%, 50%, 75% and 100% of each training fold with unchanged validation rows. Error bars show fold standard deviation. Preprocessing is fitted inside each subset. The audit contains 90 fits, including reproduced baselines; no score at 20,000 rows is extrapolated.

Full-size development condition	Accuracy	Macro F1	Emergency recall
Logistic Regression / Prior features	86.09	86.57	92.65
Logistic Regression / English details	87.40	87.83	93.38
Logistic Regression / Original details	88.99	89.38	93.63
HistGradientBoosting / Prior features	84.26	84.76	90.38
HistGradientBoosting / English details	85.54	86.02	91.48
HistGradientBoosting / Original details	87.65	88.07	92.89
Random Forest / Prior features	79.13	79.21	95.96
Random Forest / English details	79.93	80.00	96.32
Random Forest / Original details	80.24	80.32	95.89

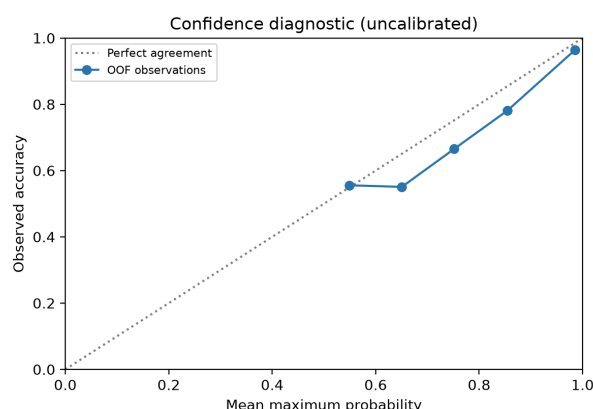
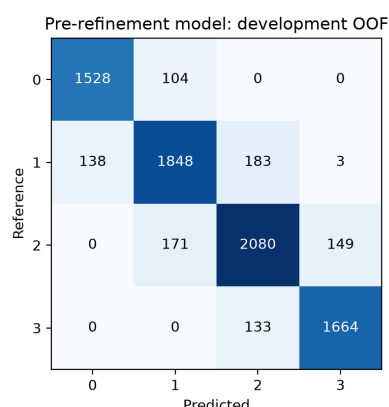
Recommended next step

Review the local error queue and duration mismatches, agree a consistent four-level rubric with qualified reviewers, and collect distinct new examples. More rows may help, but duplicated templates or unclear labels will not reliably solve the remaining Urgent/Standard errors. Reserve new independently reviewed records for confirmation before further selection.

The GUI retains the original complaint for the 19 detail features and uses checked English for SapBERT. Similarity and cluster views use full 768-D vectors; stop-word removal is inactive. The report measures supplied concepts plus original details, not end-to-end live translation accuracy.

Further Four-Level Error Investigation

This diagnostic uses only the 8,001 development records and the pre-refinement model. Supplied labels 0, 1, 2 and 3 remain unchanged. It is a diagnostic of model errors, not an independent assessment of clinical label correctness.



Pre-refinement development diagnostic	Value
Out-of-fold errors	881
Errors between neighbouring levels	878
Urgent / Standard confusions	354
Errors with confidence at least 90%	220
All three classifiers agree on the wrong level	345
Matthews correlation coefficient	0.8522
Multiclass log loss	0.2648
Multiclass Brier score	0.1586
Confidence calibration gap (10 bins)	0.0331

Higher MCC is better; lower log loss, Brier score and calibration gap are better. Confidence bins compare the mean maximum predicted probability with observed accuracy. These conditional development diagnostics do not measure end-to-end translation accuracy.

The follow-up freezes 13 settings before evaluation: LR regularization and balancing at PCA-128, PCA-64/256 alternatives, HGB leaf/regularization settings and RF feature sampling. All retain the 19 original-complaint details. Every setting uses five grouped folds; the original emergency-recall selection constraint remains fixed.

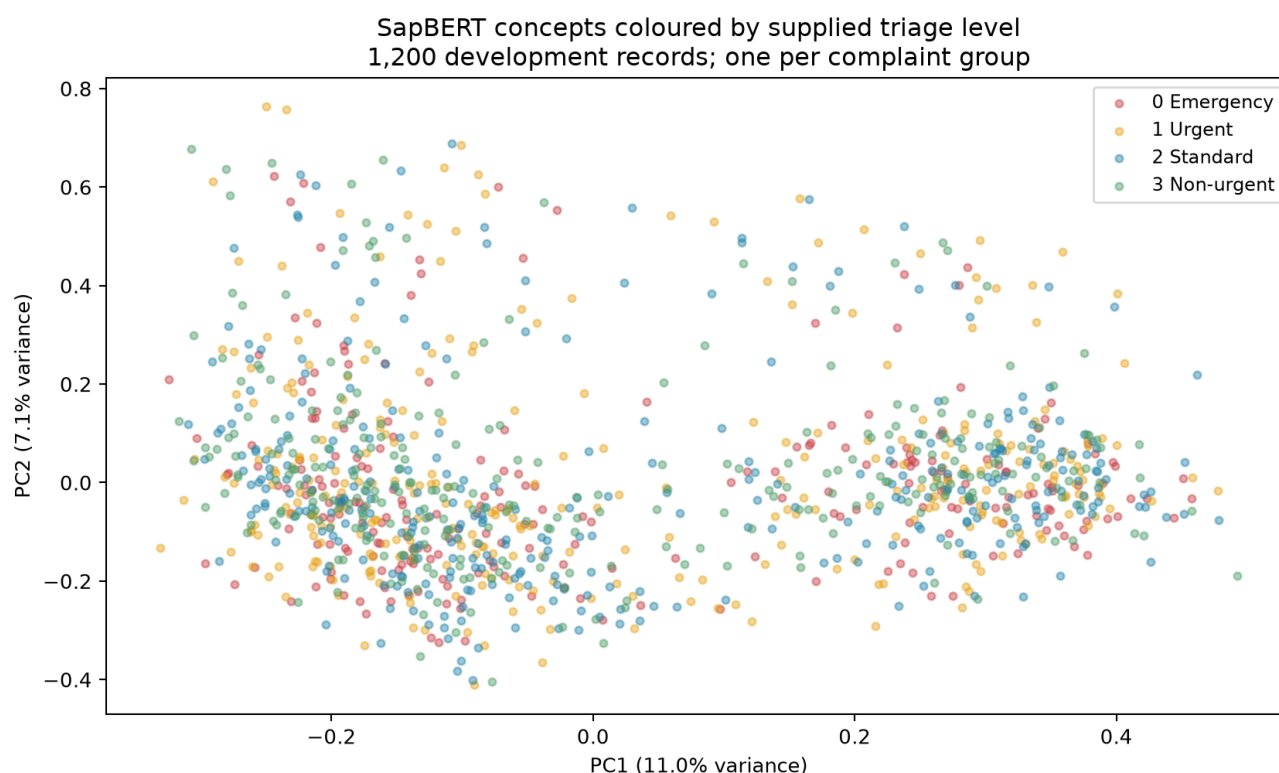
Selected development diagnostics: MCC 0.8561; log loss 0.2459; Brier score 0.1523; calibration gap 0.0120.

The older three-level scores concern a different labelling task. No labels are replaced, no difficult records are dropped and no test-based selection is used.

Historical 13-setting refinement versus its prior model: pooled development F1 gain +0.26 percentage points; conditional paired group interval -0.08 to +0.64. The interval includes zero, so a reliable gain is not established. Repeated-selection uncertainty is not included.

SapBERT Embedding Geometry

A fixed stratified sample contains 1,200 development records, one per complaint group. This diagnostic compares triage-label similarity in frozen full-768 embeddings and the fitted PCA projection. No test rows participate.



Representation	Within-label	Between-label	Difference	Silhouette
SapBERT 768-D	0.4991	0.4922	0.0069	-0.0122
PCA-64	0.0171	-0.0019	0.0190	0.0017
PCA-128	0.0163	-0.0018	0.0182	0.0016

Within-label and between-label values are mean pairwise cosine similarities. Their difference describes separation by the supplied triage labels. Silhouette near zero indicates overlapping labels in that representation. PCA centres the vectors, so absolute cosine values across raw and projected spaces are not directly interchangeable.

The low triage-label separation helps explain why a different classifier alone may give limited gains on complaint concepts. SapBERT encodes medical meaning; similar medical complaints can receive different triage levels when patient measurements and symptom details differ. The deployed classifier uses those additional features.

The plot shows only two principal components. The diagnostic uses a fixed sample and a PCA fitted on all development rows; it is descriptive, not out-of-fold predictive performance or evidence of a maximum achievable accuracy. It does not establish that any supplied label is wrong.

Additional Classifiers and the 90% Target

Eleven new settings were declared before fitting and evaluated across the same five grouped development folds: 55 additional fits. CatBoost, XGBoost and RBF SVM were compared with the previous 68 configurations. Dataset, labels, encoder and test membership remain unchanged.

New configuration	Accuracy	Precision	Recall	F1	ER recall
cat_detail128_depth4	85.56	86.01	86.26	85.94	94.24
cat_detail128_depth6	85.88	86.38	86.52	86.30	93.81
cat_detail64_depth4	86.51	86.90	87.14	86.86	94.24
cat_detail64_depth6	86.88	87.30	87.39	87.23	93.26
svc_detail128_c1	84.70	85.33	85.11	85.20	88.66
svc_detail128_c10	85.04	85.59	85.48	85.49	89.22
svc_detail128_c100	83.74	84.46	84.12	84.24	87.07
xgb_detail128_depth4	85.36	85.86	86.05	85.85	92.89
xgb_detail128_depth6	84.98	85.57	85.57	85.42	92.16
xgb_detail64_depth4	85.95	86.43	86.51	86.39	91.85
xgb_detail64_depth6	85.50	86.05	85.99	85.91	91.18

Metric	CV mean (%)	Retrospective (%)	Target (%)
Accuracy	89.28	89.29	90.00
Macro precision	89.59	89.67	90.00
Macro recall	89.78	89.81	90.00
Macro F1	89.65	89.73	90.00

The 90% target was not reached across all four aggregate metrics. Adding these classifier families did not establish the requested result. The best eligible development model is retained; no result is rounded up to imply success.

The constraint requires mean emergency recall of at least 92.38%. Selection ranks eligible settings by macro F1, then accuracy. The desired 90% aggregate target does not override this rule.

The selected configuration is unchanged from the previous round. Therefore, the deployed model and its measured scores remain unchanged; the new work expands the evidence rather than claiming a gain.

These scores measure agreement with the supplied four-level labels. They do not establish a maximum achievable score or prove that a label is wrong. The three-level study had different targets, so its higher scores are not a valid requirement for this four-level experiment.